

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.), Part-IV, Odd Semester Examination, 2024
Course: CSE4121 (Object Oriented Design and Design Patterns)
Session: 2020-2021, 2019-2020 and 2018-2019
Time: 02 hours Full marks: 35

Answer **Four** questions taking any **Two** questions from each section

Section-A

1. (a) Explain the benefit of representing real-world entities—such as *Citizen*, *MunicipalService*, and *Complaint*—in the Dhaka Urban Management System as objects in your code. 3
(b) Consider a **Dhaka Urban Management System** scenario: a city resident logs into the smart city portal using their ID and password to verify their identity with the municipal database. The user is then provided options to **report waste collection issues**, **pay utility bills**, **request road maintenance**, or **track service status**. Explain how **CRC (Class-Responsibility-Collaborator)** cards can be utilized to isolate and define the system components—including their attributes, functionalities, and interactions—for example, classes such as *Citizen*, *MunicipalService*, *Complaint*, and *Payment* can be defined along with their responsibilities and collaborating relationships. 4.75
(c) What is the difference between software design and software architecture? 1
2. (a) The design principle of '**decomposition**' takes a whole thing and divides it into different parts. There are three types of relationships in decomposition, which define the interaction between the whole and the parts: **association**, **aggregation** and **composition**. Explain those relationships and draw the UML diagram for each of these relationships based on the Dhaka Urban Management System concept. 4.75
(b) The computation steps taken within a class never need to be known by any other class, as long as they are able to access the interface. What type of encapsulation occurs here? 1
(c) In the case of an **Agricultural Automation System**, design a parent class **FarmingMachine**, and child classes **IrrigationSystem** and **Harvester**—where the child classes inherit attributes and behaviours from the parent class. Now, explain what type of **design principle** is used here. 3
3. (a) Consider the following code snippet: 3.5

```
public class OrderProcessor {  
    ...  
    public void process(Order order) {  
        order.calculateTotal();  
        order.applyDiscount();  
        order.generateInvoice();  
    }  
}
```

Identify the code smell present in this example. Describe why it is a problem and how you would refactor the code to improve the design.

(b) What is the Single Responsibility Principle (SRP)? Explain with an example showing violation and correct implementation. 3.5
(c) How can comments be considered a code smell? Provide examples of when comments indicate poor code quality. 1.75

Section -B

4. (a) What are software design patterns? Explain why learning and applying design patterns is beneficial for software developers. 2.75
- (b) Consider the following ConfigurationManager class: 4

```
class ConfigurationManager {
    private String configFilePath;

    public ConfigurationManager(String configFilePath) {
        this.configFilePath = configFilePath;
    }

    public String getProperty(String key) {
        System.out.println("Retrieved property: " + key +
                           " from " + configFilePath);
        return "value";
    }
}
```

Convert this class into a **Singleton pattern** implementation. Explain why Singleton pattern is appropriate for a configuration manager. Also, demonstrate how to use this singleton class with sample client code.

- (c) Explain the Adapter design pattern with a real-world example. 2
5. (a) Consider the activity of changing TV channels using a remote. 5
- A user can press corresponding channel number on the remote to change channel.
 - The User can also press up/down button on the remote to change the channel.

Based on the description above, draw a UML sequence diagram to model the interaction between the user and the TV through the remote.

- (b) What do you understand by conceptual integrity? Explain the ways to achieve conceptual integrity during the design and development of software. 3.75
6. (a) In Facebook, a user can follow another user. Whenever the followed user posts a new status update, all followers receive a notification in their feeds. 4.5

Which design pattern is most suitable for implementing this notification system? Explain it with a UML class diagram.

- (b) You are building a simple music player that can play only MP3 files. Later, you want the player to support other formats such as MP4 and VLC. The problem is that the library required for MP4 and VLC files works differently from your existing player and does not match the methods your system already uses. You cannot modify the library, and you do not want to rewrite your entire player. 4.25

Which design pattern can help solve this problem? Explain with an example code snippet.

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-4 Odd Semester Examination-2023
Course: CSE4121 (Object Oriented Design and Design Patterns)

Full Marks: 35

Duration: 2 Hours

[Answer any FOUR questions taking TWO from each section]

SECTION: A

1. a) Explain the benefit of representing real-world entities and in your code as object? 3
 b) Consider an ATM machine scenario: a user inserts their ATM card, enters a PIN to verify their identity with the bank, and is then given options to deposit funds, withdraw cash, or check their account balance. Explain how CRC (Class-Responsibility-Collaborator) cards can be utilized to isolate and define the system components, including their attributes, functionalities, and interactions. 4.75
 c) What is the difference between software design and software architecture? 1
2. a) The design principle of decomposition involves breaking a whole system into smaller, manageable parts. In object-oriented design, the relationships between the whole and its parts are defined by three key types: association, aggregation, and composition. Discuss these three relationships using real-world examples. Draw the corresponding UML diagram for each relationship, clearly labelling the parts and the whole. 6
 b) Consider the following Java implementation of **Whale** class inherited from **Animal**. Can you identify any problem with this implementation? Explain. 2.75

<pre> 1 public class Animal { 2 private int numberOfLegs; 3 private boolean hasTail; 4 5 public Animal(int legs, boolean tail) { 6 this.numberOfLegs = legs; 7 this.hasTail = tail; 8 } 9 10 public void walk() {.. } 11 public void run() {.. } 12 public void eat() {... } 13 }</pre>	<pre> 1 public class Whale extends Animal { 2 public Whale () { 3 super(0, true); 4 } 5 6 private void swim{ ... } 7 8 public void run() { 9 this.swim(); 10 } 11 12 public void walk() { 13 this.swim(); 14 } 15 }</pre>
--	---

3. Suppose we have a class called **Sensor** that has two purposes, getting humidity and getting temperature sensor readings. Here we have a get method that takes a zero (0) flag if you wanted to return the humidity value, and takes the one (1) flag if you want it to return the temperature value. The class diagram for the Sensor and implementation of its' **get()** method is as below:

 <pre> classDiagram class Sensor { +get(controlFlag: int) }</pre>	<pre> 1 public void get(int controlFlag) { 2 switch (controlFlag) { 3 case 0: 4 return this.humidity; 5 6 case 1: 7 return this.temperature; 8 9 default: 10 throw new UnknownControlFlagException(); 11 } 12 }</pre>
--	---

- a) Evaluate the above design in terms of coupling and cohesion. Give justification in each case. 5
- b) Suggest a better design for above sensor that meets acceptable coupling and cohesion requirement. 3.75

SECTION: B

4. a) What are design patterns, and how do they help solve common design issues in object-oriented software? 2
b) What are creational design patterns, and why are they important? Give an example of each one. 3
c) How do structural design patterns contribute to the organization of software? Explain with an example. 3.75
5. a) What is software design pattern and why should we learn them? 1.75
b) Define creational design patterns and explain their primary purpose in software development. 3
c) A software system requires a single shared instance of a logging service to record errors across multiple modules. Which creational design pattern should be used? Explain why and outline its structure. 4
6. a) What problem does Adapter design pattern solve? Explain with an example and draw supporting UML diagram. 4.5
b) Suppose You are designing a system for money withdrawal from bank account using check with the following authorization scenario: 4.25
 - For Tk. < 10,000 only cashier's authorization is sufficient to withdraw money from account.
 - For Tk. 10,000 to 10,00,000 authorization from both cashier and senior officer is needed
 - For Tk > 10,00,000 authorization from Senior officer and Manager is need.

Implement this behaviour using appropriate software design pattern

**SECTION A****[ANSWER ANY TWO OF THE FOLLOWING]**

- 1.(a) What do you understand by Object Oriented Modelling? 1.75
- (b) What is the key difference between software design and software architecture? 2.00
- (c) Explain how CSR cards are used in modelling a conceptual design with an example. What are the advantages of using CRC cards? 3+2
- 2.(a) The design principle of '**decomposition**' takes a whole thing and divides it into different parts. There are three types of relationships in decomposition, which define the interaction between the whole and the parts: **association**, **aggregation** and **composition**. Explain those relationships and also draw the UML diagram for each of these relationships. 5.00
- (b) Consider an activity of "**changing TV channel using remote**". 3.75
- When a viewer presses a number on the tv remote, the television changes the channel that the tv viewer can see on the tv screen.
 - The viewer can also press the up/down button on the remote to change the channel
- Use the above description to draw a UML sequence diagram to model the interaction between the user and TV through the remote.
- 3.(a) What do you understand by Conceptual Integrity? 1.00
- (b) Consider the following Smartphone class 2.75

```
public class SmartPhone {  
    private byte camera;  
    private byte phone;  
  
    public SmartPhone() { ... }  
  
    public void takePhoto() { ... }  
    public void savePhoto() { ... }  
    public void cameraFlash() { ... }  
  
    public void makePhoneCall() { ... }  
    public void encryptOutgoingSound() { ... }  
    public void decipherIncomingSound() { ... }  
}
```

Explain the problems with the class in terms of coupling and cohesion.

- (c) How can you apply the **separation of concern** principle to overcome the problems associated with the previous example? Draw a UML diagram of your proposed solution. 5.00

SECTION B
[ANSWER ANY TWO OF THE FOLLOWING]

- 4.(a) What are “software design patterns”, and why should you use them? 2.25
- (b) Consider the java class below. Modify the class to convert it to a singleton class. 3.50
Also, show how you can call the *executeSql* method of this singleton class using a demo program.

```
class Database {  
    private String dbName, user, password;  
  
    public Database(String dbName, String user, String password) {  
        this.dbName = dbName;  
        this.user = user;  
        this.password = password;  
    }  
  
    public void executeSql(String sql){  
        System.out.println("Executed query: "+sql);  
    }  
}
```

- (c) Describe the Adapter pattern and draw its representative class diagram. 3.00
- 5.(a) Explain how the MVC design pattern can be used to develop a maintainable web application. 4.25
- (b) What is the underlying principle of the Observer design pattern? Explain with an example. 4.50
- 6.(a) Present a scenario where you may need to use a façade design pattern. 1.75
- (b) Explain steps to implement façade design pattern 4.00
- (c) What are the facilities the façade design pattern provides? 3.00

University of Rajshahi

Department of Computer Science and Engineering

B. Sc. (Engg) Part-II Even Semester Examination 2020

Course: CSE 2211 (Object Oriented Programming and Design Methods)

[Session: 2016-2017]

Full Marks: 52.5

Duration: 3 (Three) Hours

Answer 06 (Six) questions taking any 03 (Three) from each section

Section-A

1. a) Compare Java with C++ as Object Oriented programming language. 2.00
b) Consider the following java program: 5.00

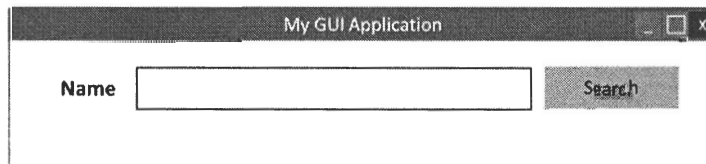
```
1 public interface IRunnable {  
2     public void run();  
3 }  
4  
5 Class JavaApp extends ClassA, ClassB implements IRunnable {  
6     void main(String *args) {  
7         System.in.println("This is a Java program.");  
8     }  
9 }
```

Point out the mistakes. Also, correct the program so that it can be compiled.

c) Mention some differences between array type and **ArrayList** type in java. 1.75
2. a) Write a method called 3.00
 `delete (String str, int m)`
 that returns the input string with the m-th element removed.
b) An n-digit number that is the sum of the n-th powers of its digits is called an Armstrong Number. For example, 153 and 370 are Armstrong numbers, since $153 = 1^3 + 5^3 + 3^3$ and $370 = 3^3 + 7^3 + 0^3$. But 15 and 37 are not Armstrong numbers, since $15 \neq 1^2 + 5^2$ and $37 \neq 3^2 + 7^2$. First few Armstrong numbers are 1, 2, 3, 4, 5, 6, 7, 8, 9, 153, 370, 371, 407, 1634 and so on. Write a Java program to find whether the given number is Armstrong or not. 4.50
c) What are the advantages of java generic classes? 1.25
3. a) What are the uses of **super** keyword in java? 1.75
b) What do you understand by 'Dynamic method dispatching'? 2.00
c) What are the conditions for a class to become abstract? Can you declare an object of abstract class? 2.00
d) Briefly describe the Object class in java. 3.00
4. a) Discuss how **try-catch-finally** block works in java. 3.00
b) In which cases **FileNotFoundException** is fired by java runtime? 1.25
c) How can you define your own exception type in java? 2.50
d) What happens when a class member is declared as static? What restrictions are applied to a method that is declared as static? 2.00

Section-B

5. a) What is thread? Discuss Java thread model. 3.25
b) Write a program to demonstrate the creation and execution of threads. How can you ensure all threads that started from main must end in order in which they started and main should end in last? 2.50
c) What are Java packages? Write the significance of packages. 3.00
6. a) What are similarities and differences between java class and java interface? 3.25
b) How java garbage collection works? What is finalization? 2.50
c) Discuss various ways java implements polymorphism. 3.00
7. a) What is the purposes of **File** class in java? 2.25
b) What are the differences between **InputStream** class and **Reader** class? 3.00
c) Write a simple java program to copy content of a text file into another file. 3.50
8. a) Discuss delegate event model with proper diagram. 3.00
b) what are the advantages of swing over AWT? 2.00
c) Write an AWT based application that uses **FlowLayout** manager to layout the following UI 3.75
that can be closed by clicking cross [x] icon in the top right corner.



University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part - II, Semester – Even, Examination 2019 (Session: 2016-2017)
Course: CSE-2211 (Object Oriented Programming and Design Methods)
Marks: 52.5 Duration: 3 hours
[N.B. Answer **THREE** questions from each section]

Property of Seminar Library
Dept. of Computer Science &
Engineering
University of Rajshahi

Section –A

1. a) What is meant by Object Oriented Programming (OOP)? Write the basic features of OOP. 1.75
b) Explain the basic structure of Java Program with example. 5
c) Why is main() method static in Java? Explain. 2
2. a) How Java garbage collection works? What is finalization? 3.5
b) Using example Java code, distinguish between overloading and overriding. 1.75
c) Briefly describe how access to member variables and methods are controlled in Java class. 3.5
3. a) What do you understand by object-oriented design pattern? Why should we use them? 2.75
b) Suppose that your program has a class that you want to have only one instance created and this instance must be reused during the lifetime of your program. Which design pattern will you use to implement that. Explain with example. 3
c) Briefly discuss the idea of Observer pattern with example. 3
4. a) What are the differences and similarities between abstract class and Interface? Mention a situation when the use of one of them is preferable over the other. 3.5
b) See the program below. Identify and explain the problems in the code. After fixing the problem, what will be the output? 3

```
class Base {
2   final public void Print() {
3       System.out.println("Base");
4   }
5 }

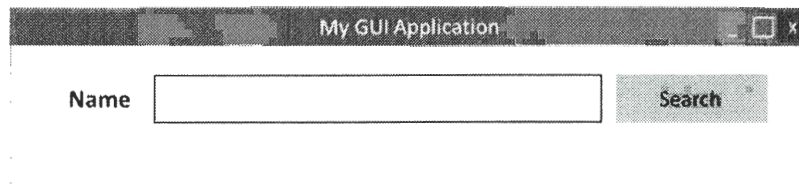
class Derived extends Base {
    public void Print() {
6       System.out.println("Derived");
7   }
8 }

9 class Main{
10     public static void DoPrint( Base o ) {
11         o.Print();
12     }
13     public static void main(String[] args) {
14         Base x = new Base();
15         Base y = new Derived();
16         Derived z = new Derived();
17         DoPrint(x);
18         DoPrint(y);
19         DoPrint(z);
20     }
21 }
```

- c) What are the purposes of **super** keyword in java? 2.25

Section –B

- 5 a) Discuss the basic structure of exception handling in Java. 3.5
b) List some common types of exceptions that may occur in Java. 1.75
c) How can you create your own type of exception? Give example 3.5
6. a) briefly explain thread model of Java. 2.75
b) How can you create a thread using runnable interface? 3
c) How does interthread communication is performed in Java? 3
7. a) Discuss delegate event model with proper diagram. 3
b) what are the advantages of swing over AWT? 2
c) Write an AWT based application that uses **FlowLayout** manager to layout the following UI that can be closed by clicking cross [x] icon in the top right corner. 3.75



8. a) What is Java package? Why it is useful? 2
b) Explain what problem can arise while using the following generic class. How to fix it? 3

```
1 class Stats<T> {  
2     T[] nums;  
3  
4     Stats(T[] o) {  
5         nums = o;  
6     }  
7  
8     double sum() {  
9         double x = 0.0;  
10        for(int i=0; i < nums.length; i++)  
11            x += nums[i].doubleValue();  
12  
13        return x;  
14    }  
15 }
```

- c) Write a java program that reads the content of a file and writes the content back to another file. 3.75

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-2 Even Semester Examination-2018
Course: CSE2211 (Object Oriented Programming and Design Methods)
Marks: 52.50 Time: 3 Hours

[N.B. Answer SIX questions taking THREE from each section.]

Section-A

1. a) Explain the differences between 2, 2.0, '2', "2", and "2.0" values of java variable. 2.25
- b) Find three *run-time* errors in the following program. 2.25
- ```
1. public class HasErrors {
2. public static void main(String[] args) {
3. int x = 0;
4. int y = 0;
5. Scanner in = new Scanner("System.in");
6. System.out.print("enter an integer:");
7. x = in.readInt();
8. System.out.print("enter another integer: ");
9. x = in.readInt();
10. System.out.println("The sum is " + x + y);
11. }
12. }
```
- c) The volume of a sphere is given by the formula  $V = \frac{4}{3}\pi r^3$ . If the radius is given by a variable radius of type double, write a Java expression for the volume. 1.25
- d) Suppose the value of b is false and the value of x is 0. What is the value of each of the following expressions? 2
- i. !b && x == 0
  - ii. b || x != 0
  - iii. !b && x != 0
  - iv. !b || x != 0
- e) What is the value of mystery after the following sequence of statements? 1
- ```
int mystery = 1;
mystery = 1 - 2 * mystery++;
mystery = mystery + 1;
```
2. a) Write a program that reads a number between 1 and 999,999 from the user, where the user enters a comma in the input when and where needed. Then, always print the number without a comma. Here are some sample I/O: 3.25
- Input:* Please enter an integer between 1 and 999,999: 23,456
Output: 23456
Input: Please enter an integer between 1 and 999,999: 123
Output: 123
- b) Explain the difference between 2
- ```
s = 0;
if (x > 0) { s++; }
if (y > 0) { s++; }
```
- and
- ```
s = 0;
if (x > 0) { s++; }
else if (y > 0) { s++; }
```

c) Write a program using loops that compute the sum of all odd digits of an input. (For example, if the input is 32677, the sum would be $3 + 7 + 7 = 17$.) 3.50

3. a) Why is the **enhanced for loop** not an appropriate shortcut for the following **basic for loop**? 1.75

```
for (int i = 0; i < values.length; i++) {  
    values[i] = i * i;  
}
```

b) What is wrong with each of the following code segments? 4

i.

```
int[] values = new int[10];  
for (int i = 1; i <= 10; i++) {  
    values[i] = i * i;  
}
```

ii.

```
int[] values;  
for (int i = 0; i < values.length; i++) {  
    values[i] = i * i;  
}
```

iii.

```
ArrayList<Integer> values = new ArrayList<Integer>();  
for (int i = 1; i <= 10; i++){  
    values.set(i - 1, i * i);  
}
```

iv.

```
ArrayList<Integer> values;  
for (int i = 1; i <= 10; i++) {  
    values.add(i * i);  
}
```

c) Assuming `SavingsAccount` is a subclass of `BankAccount`, which of the following code fragments are valid in Java? 2

- i. `BankAccount account = new SavingsAccount();`
- ii. `SavingsAccount account2 = new BankAccount();`
- iii. `BankAccount account = null;`
- iv. `SavingsAccount account2 = account;`

d) Consider this top-level and inner class. Which variables can the `f()` method access? 1

```
public class T {  
    private int t;  
    public void m(final int x, int y) {  
        int a;  
        final int b;  
        class C implements I {  
            public void f() {  
                ...  
            }  
        }  
        final int c;  
        ...  
    }  
}
```

4. a) Explain the output of below code snippet? 1.75

```
String s1 = new String("abc");  
String s2 = new String("abc");  
System.out.println(s1 == s2);
```

b) What are the differences between method overloading & method overriding? Explain with short examples. 4

c) Once created, Java String objects are immutable. But how would you create a string that can be changed? 3
Explain with example.

Section-B

5. a) If a program Woozle is started with the command
`java Woozle -Dname=piglet -I\eeeyore -v heff.txt a.txt lump.txt`
what are the values of `args[0]`, `args[1]`, and so on? 2
- b) The following code tries to close the writer without using a try-with-resources statement:
`PrintWriter out = new PrintWriter(filename);
try{
 Write output. out.close();
}catch (IOException exception){
 out.close();
}`
What is the disadvantage of this approach? 2
- c) What is the error in the following code? 2.75
`class Parent{
 public void displayMsg() throws IOException{
 System.out.println("In Parent displayMsg()");
 throw new IOException("Problem in displayMsg_Parent");
 }
}
public class OverrideDemo extends Parent{
 public void displayMsg() throws Exception{
 System.out.println("In OverrideDemo displayMsg()");
 throw new Exception("Problem in displayMsg_Child");
 }
}`
- d) How can you create your own type of exception? (Coding example is required) 2
6. a) The following code is part of an airline booking system that is designed to be used in a single threaded application. 3
`public class Seat Counter{
 private int count = 0;
 public int getSeatCount(){
 return count;
 }
 public void bookSeat(){
 count++;
 }
 public void unBookSeat(){
 count--;
 }
}`
- i. Describe a potential problem with this code when used with multiple concurrent threads. Provide an example that demonstrates the problem.
- ii. Fix the code so that it works when used with multiple threads.
- b) Consider the following runnable. 2.50
`public class MyRunnable implements Runnable{
 public void run() {
 try {
 System.out.println(1);
 Thread.sleep(1000);
 System.out.println(2);
 }catch (InterruptedException exception){
 System.out.println(3);
 }
 }
}`

Section-B

5. a) If a program Woozle is started with the command
`java Woozle -Dname=piglet -I\eeeyore -v heff.txt a.txt lump.txt`
what are the values of `args[0]`, `args[1]`, and so on? 2
- b) The following code tries to close the writer without using a try-with-resources statement:
`PrintWriter out = new PrintWriter(filename);`
`try{`
 `Write output. out.close();`
`}catch (IOException exception){`
 `out.close();`
`}`
What is the disadvantage of this approach? 2
- c) What is the error in the following code? 2.75
`class Parent{`
 `public void displayMsg() throws IOException{`
 `System.out.println("In Parent displayMsg()");`
 `throw new IOException("Problem in displayMsg_Parent");`
 `}`
`}`
`public class OverrideDemo extends Parent{`
 `public void displayMsg() throws Exception{`
 `System.out.println("In OverrideDemo displayMsg()");`
 `throw new Exception("Problem in displayMsg_Child");`
 `}`
`}`
- d) How can you create your own type of exception? (Coding example is required) 2
6. a) The following code is part of an airline booking system that is designed to be used in a single threaded application. 3
`public class Seat Counter{`
 `private int count = 0;`
 `public int getSeatCount(){`
 `return count;`
 `}`
 `public void bookSeat(){`
 `count++;`
 `}`
 `public void unBookSeat(){`
 `count--;`
 `}`
`}`
- i. Describe a potential problem with this code when used with multiple concurrent threads. Provide an example that demonstrates the problem.
- ii. Fix the code so that it works when used with multiple threads.
- b) Consider the following runnable. 2.50
`public class MyRunnable implements Runnable{`
 `public void run() {`
 `try {`
 `System.out.println(1);`
 `Thread.sleep(1000);`
 `System.out.println(2);`
 `}catch (InterruptedException exception){`
 `System.out.println(3);`
 `}`
 `}`

```

        System.out.println(4);
    }
}

```

Suppose a thread with this runnable is started and immediately interrupted:

```

Thread t = new Thread(new MyRunnable());
t.start();
t.interrupt();

```

What output is produced?

- c) What is a deadlock? How can it be avoided? 2.25
 - d) Suppose you surround each call to the `sleep()` method with a try/catch block to catch an `InterruptedException` and ignore it. What problem do you create? 1
7. a) Provide a static method `void append(ArrayList<T> a, ArrayList<T> b)` that appends the elements of `b` to `a`. 1.75
- b) Modify the `append` method of Question 7(a) so that the second array list can contain elements of a subclass. For example, if `people` is an `ArrayList<Person>` and `students` is an `ArrayList<Student>`, then `append(people, students)` should compile but `append(students, people)` should not. 2
- c) Write a lambda expression for a function that turns a string into a string made of the first letter, three periods, and the last letter, such as "W...d". (Assume the string has at least two letters.) 3
- d) Assuming that `words` is a `Stream<String>`, what is the result of this call? 2
`words.filter(s -> s.equals(s.toUpperCase()))`
8. a) Write a Java code-fragment for adding two buttons to the northern position of a frame so that they are shown next to each other. 2
- b) What is the advantage of a `JComboBox` over a set of radio buttons? What is the disadvantage? 2
- c) Compare among AWT, Swing and JavaFX GUI APIs of Java. 4.75

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-2 Even Semester Examination -2017
Course: CSE2211 (Object Oriented Programming and Design Methods)
Full Marks: 52.5 Time: 3:00 Hours

[N.B. Answer any six questions taking THREE from each of the groups]

Section-A

- 1 a) How can you create an **ArrayList**? How do you add and remove elements from it? 2.75
How can you obtain an **Array** from an **ArrayList** ?

- b) Write an enhanced *for loop* (*foreach loop*) to get all the prices from the following 3
enum. Write the output of your code as well.

```
enum Apple {  
    Jonathan(10), GoldenDel, RedDel(12), Winesap(15);  
    private int price;  
    Apple(int p) { price = p; }  
    Apple() { price = -1; }  
    int getPrice() { return price; }  
}
```

- c) Try to find error(s) from the following program. If you found any error(s), correct 3
them and finally write the output of the program.

```
public class ErrorValue {  
    static double m(Double v) {  
        return v;  
    }  
    public static void main(String args[]) {  
        Double iOa = m(100.0);  
        System.out.println(iOa);  
  
        Integer iOb = 100;  
        int i = iOb;  
        System.out.println(i + "\n" + iOb);  
    }  
}
```

2. a) Trace the following code and write its output

3

```
//Using break to exit from nested loops
public class ControllStatement {
    public static void main(String args[]) {
        outer: for(int i=0; i<3; i++) {
            System.out.print("Pass " + i + ": ");
            inner: for(int j=0; j<100; j++) {
                if(j == 10) break outer;
                if(j % 2 == 0) continue inner;
                System.out.print(j + " ");
            }
            System.out.println("This will not print");
        }
        System.out.println("Loops complete.");
    }
}
```

b) Consider the following logical operation below.

3

```
int a =3;
int b =6;
int c =a | b;
int d=a & b;
int e=a ^ b;
int f=(~a & b)|(a & ~b);
int g=~a & 0x0f;
```

What are the values of c, d, e, f, and g?

```
int a =1;
int b =2;
int c =3;
int d = 7;
a |=4;
b >>=1;
c <<=1;
a ^=c;
d'>>>= 1;
```

What are the values of a, b, and c?

c) Can you pass `List<String>` to a method which accepts `List<Object>` .
How to write a generic method which accepts generic argument and return Generic Type?

2.75

3. a) What is the difference between an Interface and an Abstract class?

2

b) What is method overloading and method overriding? Give an example.

2

```

public interface Red{
    public void turnRed();
}
public interface Blue{
    public void turnBlue();
}
public class A implements Red {
    public void one() {
        System.out.println( "A.1" );
    }
    public void two(){
        one();
        System.out.println( "A.2" );
    }
    public void turnRed() {
        System.out.println("Noo!!");
    }
}
public class B extends A implements Red, Blue {
    public void one( int x){
        System.out.println( "B.1 " + x );
    }
    public void two() {
        System.out.println( "B.2" );
    }
    public void turnBlue() {
        System.out.println("Aie!!");
    }
}
public class C extends A {
    public void one() {
        System.out.println( "C.1" );
    }
}

```

For each of the following situation, indicate what output is produced when the code is executed. If some sort of error occurs instead, give a brief explanation of the problem.

- i. `A a = new B();`
`a.two();`
- ii. `B b = new B();`
`b.one(4);`
`b.one();`
- iii. `Red r = new B();`
`r.turnRed();`

```

iv.  Blue bb =new Blue();
      bb.turnBlue();

v.   C c =new A();
      c.two();

vi.  A aa =new B();
      aa.turnBlue();

vii. A ab =new C();
      ab.two();

```

4. a) Find error(s) in the following program. Suggest a way to correct the program if you found any error(s). And finally write the output of the program, if you run by the following commands

5

```

i.    java MultipleCatches
ii.   java MultipleCatches Something

```

```

1  public class MultipleCatches {
2      public static void main(String args[]) {
3          try {
4              int a = args.length;
5              System.out.println("a = " + a);
6              int b = 42 / a;
7              int c[] = { 1 };
8              c[42] = 99;
9          } catch (Exception e) {
10             System.out.println("Exception catch.");
11         } catch (ArithmeticException e) {
12             System.out.println("Divide by 0: " + e);
13         } catch (ArrayIndexOutOfBoundsException e) {
14             System.out.println("Array ioob: " + e);
15         } catch (ArrayStoreException | ClassCastException e){
16             System.out.println("Array Related " + e);
17         }
18     }
19 }

```

b) How can you create your own type of exception? (Coding example is required)

3.75

Section-B

5. a) What is the difference between `start()` and `run()` methods of Thread class? 2
b) What happens when an Exception occurs in a thread? 1
c) How do you share data between two threads in Java? 2
d) Discuss the process of creation of server and client sockets with exceptions handled explicitly with a suitable example 2
e) How do you connect to a URL resource in Java programming language? 1.75
6. a) What are the rules of defining a functional interface? 4
Rewrite the following code in lambda expression.
- i. **//Creating a new thread.**
- ```
Thread thread = new Thread(new Runnable() {
 public void run() {
 System.out.println("Hello World!");
 }
});
```
- ii. **//Handling JavaFX action events for the btnHourglass button.**
- ```
btnHourglass.setOnAction(new EventHandler<ActionEvent>() {  
    public void handle(ActionEvent ae) {  
        response.setText("Hourglass Pressed");  
    }  
});
```
- b) Carefully examine the following program and try to find error(s) if any in the code. If you found any error(s), correct them and finally write its output for the following runs 4.75
- i. `java LambdaDemo 10 2`
ii. `java LambdaDemo`
iii. `java LambdaDemo 10 3`
- ```
// LambdaDemo.java
public class LambdaDemo {
 public static void main(String args[])
 {
 int a, b;
 String factorOrNot = "";
 a = Integer.parseInt(args[0]);
 b = Integer.parseInt(args[1]);
 NumericTest isFactor = (n, d) -> (n % d) == 0;
 if(!isFactor.test(a, b))
 factorOrNot = "a factor";
 else
 factorOrNot = "not a factor";
 System.out.println(b+" is "+factorOrNot+" of "+a);
 }
}
interface NumericTest{
 boolean test(int n, int d);
}
```
7. a) Where should you write a file closing statement to close a file stream. How can you open a file stream so that it will close automatically when it is no longer in use? 3

- b) What are the differences between Byte Streams and Character Streams 1
- c) What is the difference between comparing with == operator and equals ( ) method? 2

Write the output of the following code snippet.

```
String s1 = "CSE";
String s2 = "CSE";
String s3 = new String("CSE");
System.out.println(s1 == s2);
System.out.println(s1 == s3);
System.out.println(s1.equals(s2));
System.out.println(s1.equals(s3));
System.out.println(s1.compareTo(s2));
System.out.println(s1.compareTo(s3));
```

- d) What does it mean by "String is Immutable in Java"? Why String is Immutable in Java? 2.75

Write down the output of the following code snippet.

```
String str = new String("Hello");
System.out.println(str);
str.concat(" World");
System.out.println(str);
StringBuffer sb = new StringBuffer("Hello");
System.out.println(sb);
sb.append(" World");
System.out.println(sb);
```

8. a) With suitable block diagram, explain Delegation Event Model in java 3
- b) Why is swing preferable for GUI programming over AWT? 2
- c) Write a simple JavaFX skeleton application and discuss its key components 3.75

**University of Rajshahi**  
**Department of Computer Science & Engineering**  
B.Sc. (Engg.) Part-2 (Even Semester) Examination 2016  
Course: CSE2211 (Object Oriented Programming and Design Methods)

Full Marks: 52.5

Duration: 3(Three) Hours

**Answer 06(Six) questions taking any 03(Three) questions from each part**

**Part-A**

1. a) How Java is different from other Object Oriented languages like C++? 2.5  
b) How does Java achieve its portability and strong security features? 2.5  
c) Despite being interpreted, how does Java achieve its high performance? 1.25  
d) How Java applet is different from Java application? 2.5
2. a) Explain encapsulation, inheritance and polymorphism in terms of Java programming language. 3  
b) Consider the following definition of the main method: 4.5  

```
Public static void main(String args[]) throws IOException{
 // code goes here
}
```

Explain the purpose of each keyword and identified of the above Java code segment.

c) Write down different ways of creating and initializing a 2D array in Java. 1.25
3. a) Write a Java program that accepts any arbitrary number of integers as command line argument and prints the largest number. 3  
b) Consider the following code segment that manipulates a class Box having a public variable named weight: 3  

```
Box x = new Box(); Box y = new Box(); Box z;
x.weight = 10; y.weight = 2;
z = y;
z.weight = 100;
z = x;
System.out.println(x.weight);
System.out.println(y.weight);
System.out.println(z.weight);
```

What will be the output? Justify your answer.

c) How does garbage collection in Java work? 2.75
4. a) What is an inner class? How is the scope of inner class is defined? 2.5  
b) Why is the "final" keyword used in Java? 1.25  
c) What is a constructor? Can you call one constructor from another if a class has multiple constructors? 2  
d) What is the difference between Path and Classpath? 3

### **Part-B**

- |                                                                                                 |       |
|-------------------------------------------------------------------------------------------------|-------|
| 5. a) Why should you define your classes within your own package? How can you define a package? | 1.5+1 |
| b) What are similarities and differences between Java class and Java interface?                 | 3     |
| c) How can you use interface for dynamic method dispatching?                                    | 3.25  |
| 6. a) What is a thread?                                                                         | 1     |
| b) Describe the life cycle of a thread.                                                         | 3.75  |
| c) What is the difference between processes and threads?                                        | 2     |
| d) How do you ensure that <i>n</i> threads can access <i>n</i> resources without deadlock?      | 2     |
| 7. a) What are the steps involved in event handling in Java?                                    | 2.5   |
| b) Why should you use event adapter class <b>instead</b> of event listener?                     | 2     |
| c) What do you understand by GUI programming? What is its advantage?                            | 1.25  |
| d) Write a simple Java program to show a closable graphical window.                             | 3     |
| 8. a) What is a socket?                                                                         | 1     |
| b) Explain advantages of Java Sockets.                                                          | 2     |
| c) Briefly discuss RMI.                                                                         | 4     |
| d) What is meant by binding in RMI?                                                             | 1.75  |

# University of Rajshahi

## Department of Computer Science and Engineering

B.Sc. (Engg.) Part-II (Even Semester) Examination-2015

CSE2211 (Object Oriented Programming and Design Methods)

Full Marks: 52.5

Time: 3 Hours

[Answer any six (06) questions taking three questions from each part]

### Part-A

1. (a) What are the similarities and differences between Java and C++? 2.5  
(b) What is the role of Java Virtual Machine? 2.5  
(c) Consider the following Java program written by a novice Java programmer: 3.75  

```
Class JavaTest extends ClassA, ClassB {
 public void main(String *args) {
 System.in.println("This is a simple Java program.");
 }
}
```

Point out his mistakes. Also correct the program so that it can be compiled.
2. (a) Why is Java called strangely typed language? What are rules for automatic type conversion of a variable in Java? 2.5  
(b) How can you generate the following multi-dimensional array in Java? 1.5  

|        |        |        |        |
|--------|--------|--------|--------|
| [0][0] |        |        |        |
| [1][0] | [1][1] |        |        |
| [2][0] | [2][1] | [2][2] |        |
| [3][0] | [3][1] | [3][2] | [3][3] |
- (c) How does Java garbage collection work? 1.75  
(d) What is *Instance Variable Hiding* problem? How this problem can be solved? 3
3. (a) With example explain why you would declare a variable or method as static. What are the restrictions of declaring a variable or method as static? 2+1  
(b) How can you pass command line argument to a Java console application? Explain with an example. 3  
(c) What is the difference between **String** and **StringBuffer** class? 2.75
4. (a) What are the differences between 'lightweight' and 'heavyweight' GUI components in Java? 4  
(b) What is an exception? List some common types of exception that may occur in Java. 1+3=4  
(c) In Java which collection class allows us to retrieve objects in FIFO (first-in, first-out) sequence? 0.75

### Part-B

5. (a) What is thread? Discuss Java thread model. 3.25  
(b) Discuss different ways of creating thread in Java. 3.5  
(c) How can you define a Java package? How can you use a class belonging to a different package within your package? Explain. 1+1

6. (a) Write down the differences between the following terms. 3  
 (i) Array and ArrayList (ii) method overriding and method overloading.
- (b) An  $n$ -digit number that is the sum of the  $n$ -th powers of its digits is called an  $n$  Narcissistic number. It is also sometimes known as an Armstrong number. For example, 153 and 370 are Armstrong numbers, since  $153 = 1^3 + 5^3 + 3^3$  and  $370 = 3^3 + 7^3 + 0^3$ . But 15 and 37 are not Armstrong numbers, since  $15 \neq 1^2 + 5^2$  and  $37 \neq 3^2 + 7^2$ . First few Armstrong numbers are 1, 2, 3, 4, 5, 6, 7, 8, 9, 153, 370, 371, 407, 1634 and so on. Write a Java program to find whether the given number is Armstrong or not. 5.75
7. (a) Write a Java program to reverse a string by applying your own logic without using any Java API like **StringBuffer** or **StringBuilder**. 5.75
- (b) How do you ensure that N thread can access N resources without deadlock? 2
- (c) Show two different ways of converting **String** object into **int** or **Integer** in Java. 1
8. (a) Write skeleton of a Java applet program and explain each part of the program. 3.5
- (b) Discuss how **try-catch-finally** block works in Java. 2.25
- (c) Why an interface is used in Java? How an Interface is declared and used? 1+2

**University of Rajshahi**  
**Department of Computer Science and Engineering**  
B. Sc. (Engg) Part-II Odd Semester Examination 2014  
Course: CSE-2121 (Object Oriented Programming with Java)  
Full Marks: 52.5      Duration: 3(Three) Hours

**Answer 06(Six) questions taking any 03(Three) questions from each part in separate answer script**

**Part - A**

1. a) What does Object Oriented Programming (OOP) mean? Write the applications of OOP. 2  
b) Explain the following concepts of Object Oriented Programming: 3  
(i) Data abstraction and (ii) Polymorphism  
c) Explain the steps to be followed for executing a Java program. 3  
d) What is byte-code? 0.75
2. a) What does meant by literals? Shortly explain integer, floating-point, boolean, character and string literals used in Java programming. 3  
b) Discuss the scope and lifetime of a variable. 3  
c) Show and discuss the general form of a java class. 2.75
3. a) What is class? How does it accomplish data hiding? 3  
b) In what ways do you initialize of instance fields? When do you declare a method or class final? 3  
c) What are objects? How are they created from a class? 2.75
4. a) What is a constructor? What are its special properties? 3  
b) How do we invoke a constructor? Explain with example. 3.75  
c) What are the differences between a constructor and a method? 2

**Part – B**

5. a) What is meant by inheritance and what are its advantages? Explain the use of super() with example. 2+2  
b) Explain the various form of inheritance. 3  
c) Explain why do need to use 'final' with inheritance. 1.75
6. a) Define abstract class, concrete class and interface. Explain the use of abstract class. 3+1  
b) Give the syntax of interface. Differentiate between overriding and overloading. 2.75  
c) What are wrapper classes? Why should we need a wrapper class? 2
7. a) Explain the following Java keywords with appropriate example. 5  
i. try, ii. catch, iii. throw, iv. throws v. finally.  
b) Define multithreaded programming? 2  
c) What is inner class? Give example. 1.75
8. a) What are the differences between Java Applet Web Application and JavaScript? 2  
b) Define Java Bean? What are the advantages of Java Beans? 2.50  
c) Distinguish between Abstract Window Toolkit (AWT) components and Java Swing components. 2  
d) Write a java program to display a file image on web-browser. 2.25